रा.इ.सू.प्रौ.सं
NIELIT
BHUBANESWAR

# INTRODUCTION TO ARTIFICIAL INTELLIGENCE

# Contents


- Introduction to Artificial Intelligence
- Evolution of AI
- Types of Artificial Intelligence
- Types of data in Artificial Intelligence
- AI vs Machine Learning vs Deep Learning
- Applications of Artificial Intelligence
- Advantages and Challenges of AI
- Future of Artificial Intelligence
- Exercise: Identify AI vs non-AI applications

# Introduction to Artificial Intelligence

# Definition of Artificial Intelligence

"AI is the science of making machines do things that would require intelligence if done by men." — ***Marvin Minsky***

- Artificial Intelligence enables machines to simulate human cognitive abilities — including learning, reasoning, problem-solving, and decision-making.
- AI systems can analyze data, recognize patterns, and improve over time.
- Example: Chatbots, voice assistants, recommendation systems.

# Scope of AI

AI is a broad field that spans multiple disciplines and subfields:

- Machine Learning — systems that learn from historical data
- Natural Language Processing (NLP) — understanding and generating human language
- Computer Vision — interpreting images and video
- Robotics — physical agents that sense and act
- Expert Systems — rule-based decision making
- Generative AI — creating text, images, code, and audio

AI intersects with mathematics, statistics, neuroscience, linguistics, and software engineering.

# Importance of AI in modern world

AI is no longer experimental — it powers the global economy and critical infrastructure:

- Drives innovation in healthcare, finance, education, and defense
- Enables automation of repetitive and high-risk tasks
- Processes data at a scale impossible for humans
- Powers personalized experiences at billions of touchpoints daily
- Estimated to add **$15.7 trillion** to the global economy by 2030 (PwC)

Nations and companies that lead in AI will lead the future.

# Everyday impact of AI

- Voice assistants like Alexa and Google Assistant.
- Recommendation systems in Netflix, YouTube, Amazon.
- Face recognition in smartphones.
- Spam filtering in emails.
- Navigation apps like Google Maps providing real-time traffic updates.

# Evolution of AI

# Early AI (Rule-Based Systems)

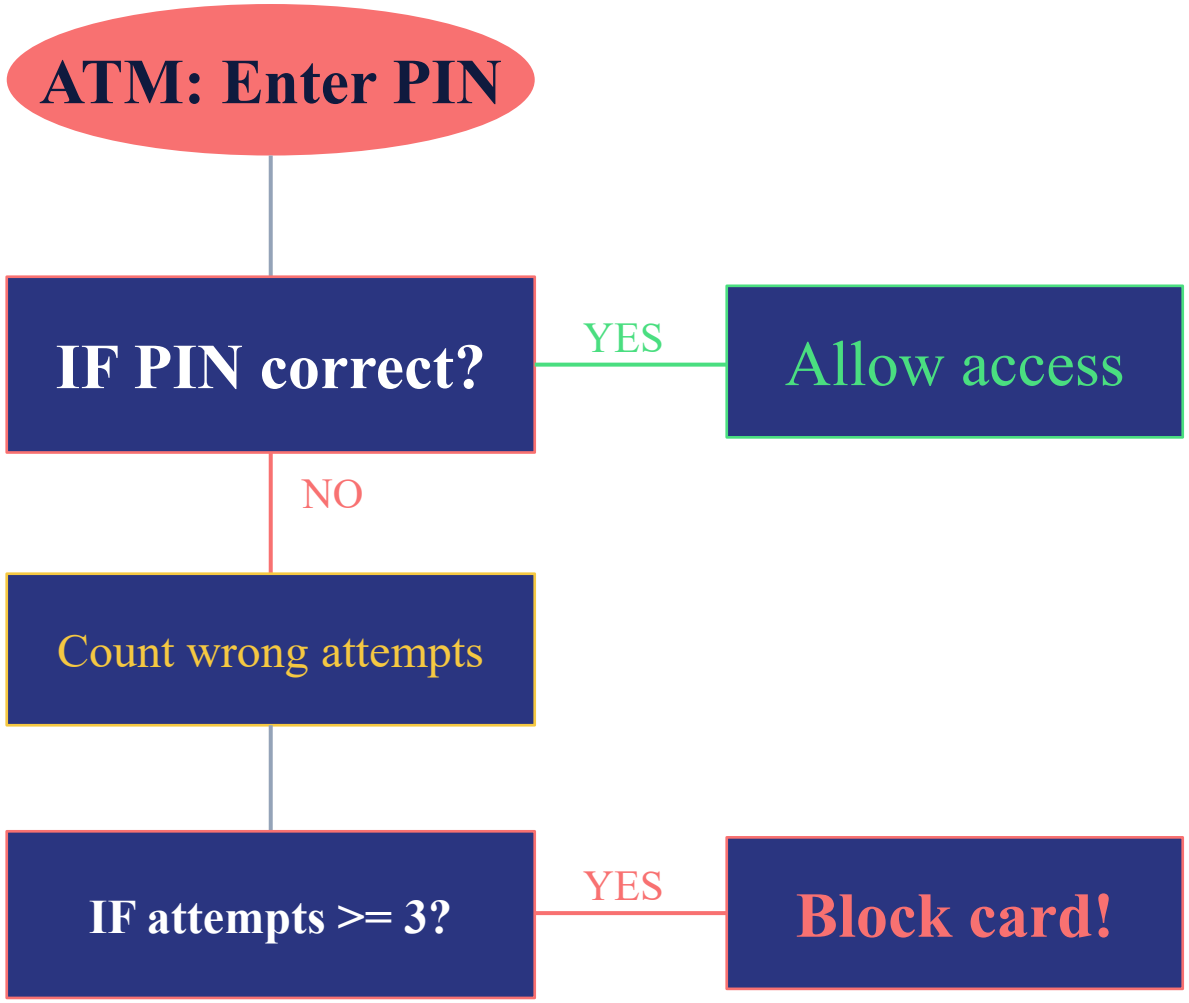
The first AI systems didn't learn — they followed strict rules written by humans.

- Programs were built using if-then logic: "**IF temperature > 100 THEN alert**"
- Called **Expert Systems** — encoded the knowledge of human experts into rules
- Famous example: **MYCIN (1970s)** — diagnosed blood infections using 600 rules
- **ELIZA (1966)** — simulated conversation using pattern matching

Limitation: couldn't handle situations the programmer didn't anticipate. Rule-based AI is like a recipe. It works perfectly — until an ingredient is missing.

**Human Knowledge → Rules Written in Code → Machine Follows Rules → Output**

# How It Works — IF / THEN / ELSE Logic




## Real Examples

**ATM PIN check**

*IF pin_wrong >= 3 → block card*

**Calculator**

*IF operator = '+' → add numbers*

**Traffic light**

*IF timer = 0 → switch signal*

**Exam eligibility**

*IF attendance < 75% → bar student*

# Rise of Machine Learning

- Machine Learning (ML) allows machines to learn from data instead of being explicitly programmed.
- Systems improve their performance with experience.
- Common types: Supervised, Unsupervised, and Reinforcement Learning.
- Used in spam detection, recommendation systems, and fraud detection.
- Advantage: More flexible and adaptive than rule-based systems.

# Emergence of Deep Learning

- A breakthrough that made machines see, hear, and understand language.
  - Inspired by the structure of the human brain — **artificial neural networks**
  - **Deep** = many layers of neurons stacked together
  - **2012 turning point** — AlexNet wins ImageNet with 85% accuracy, shocking the world
  - Powered by three forces: **Big Data + GPUs + Better algorithms**
- Key architectures:
  - **CNN** — for images (face recognition, medical imaging)
  - **RNN/LSTM** — for sequences (speech, text)
  - **Transformer (2017)** — foundation of modern AI

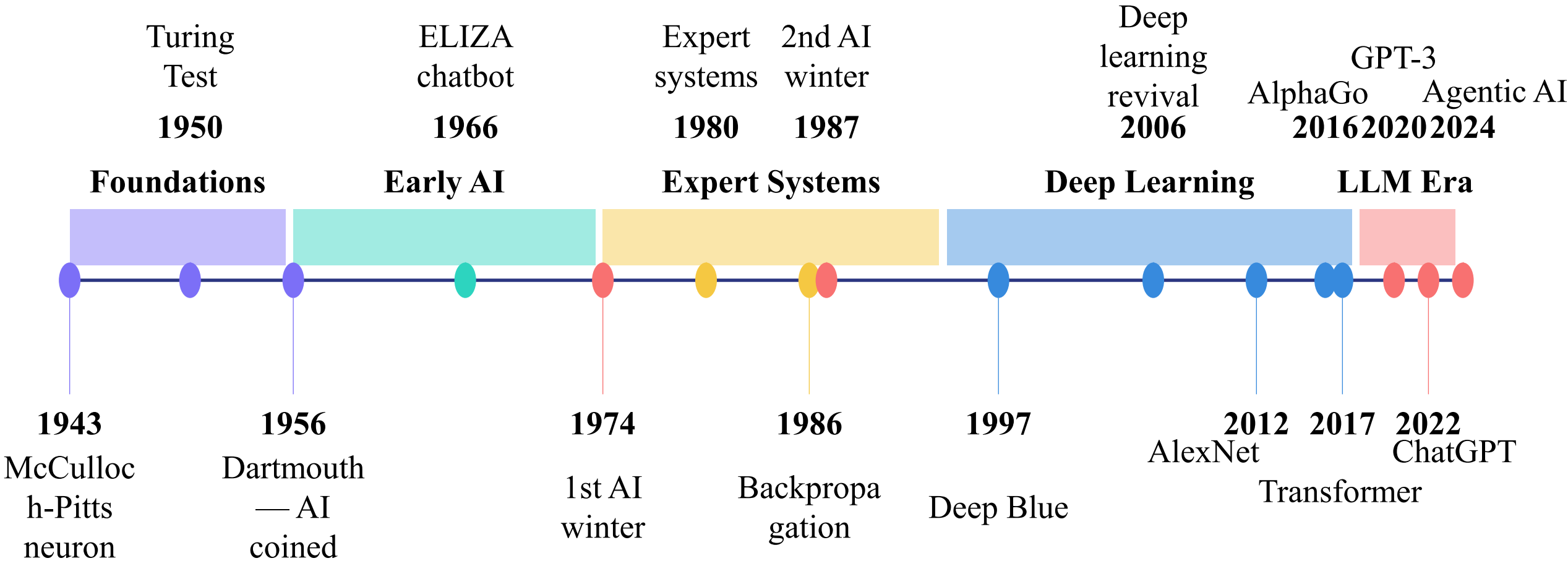
# Growth of Generative AI

AI that doesn't just analyze — it creates.

- Generative AI produces new content: text, images, audio, video, code
- Powered by Large Language Models (LLMs) trained on trillions of words

- Key milestones:
  - GPT-3 (2020) — human-like text generation
  - DALL-E / Stable Diffusion (2021–22) — text to image
  - ChatGPT (2022) — 100 million users in 2 months
  - GPT-4, Gemini, Claude (2023–24) — multimodal reasoning
- Applications: content creation, coding assistants, drug discovery, education

# Timeline of major developments



# Types of Artificial Intelligence

# Types of AI

WE ARE HERE

*Where are we today?*

**Narrow AI** → **General AI** → **Super AI**

## Narrow AI

Performs ONE specific task. All current AI systems.

*Siri · ChatGPT · Face ID · Google Maps*

**EXISTS TODAY**

## General AI

Performs ANY task a human can. Learns across domains.

*No real example yet*

**RESEARCH STAGE**

## Super AI

Surpasses human intelligence in every field.

**THEORETICAL**

*Narrow AI is the ONLY type that exists today. General AI and Super AI remain future goals.*

# Types of data in Artificial Intelligence

# Understanding data

- Data is the backbone of AI.
- It's like a brain's memories.
- Without data, AI can't recognize patterns or make decisions.
- For example: An AI system that suggests music needs to know your favourite genres first.

# Types of data

**Unstructured data**

The university has 5600 students.
John's ID is number 1, he is 18 years old and already holds a B.Sc. degree.
David's ID is number 2, he is 31 years old and holds a Ph.D. degree. Robert's ID is number 3, he is 51 years old and also holds the same degree as David, a Ph.D. degree.

**Semi-structured data**

```
<University>
  <Student ID="1">
    <Name>John</Name>
    <Age>18</Age>
    <Degree>B.Sc.</Degree>
  </Student>
  <Student ID="2">
    <Name>David</Name>
    <Age>31</Age>
    <Degree>Ph.D. </Degree>
  </Student>
....
</University>
```

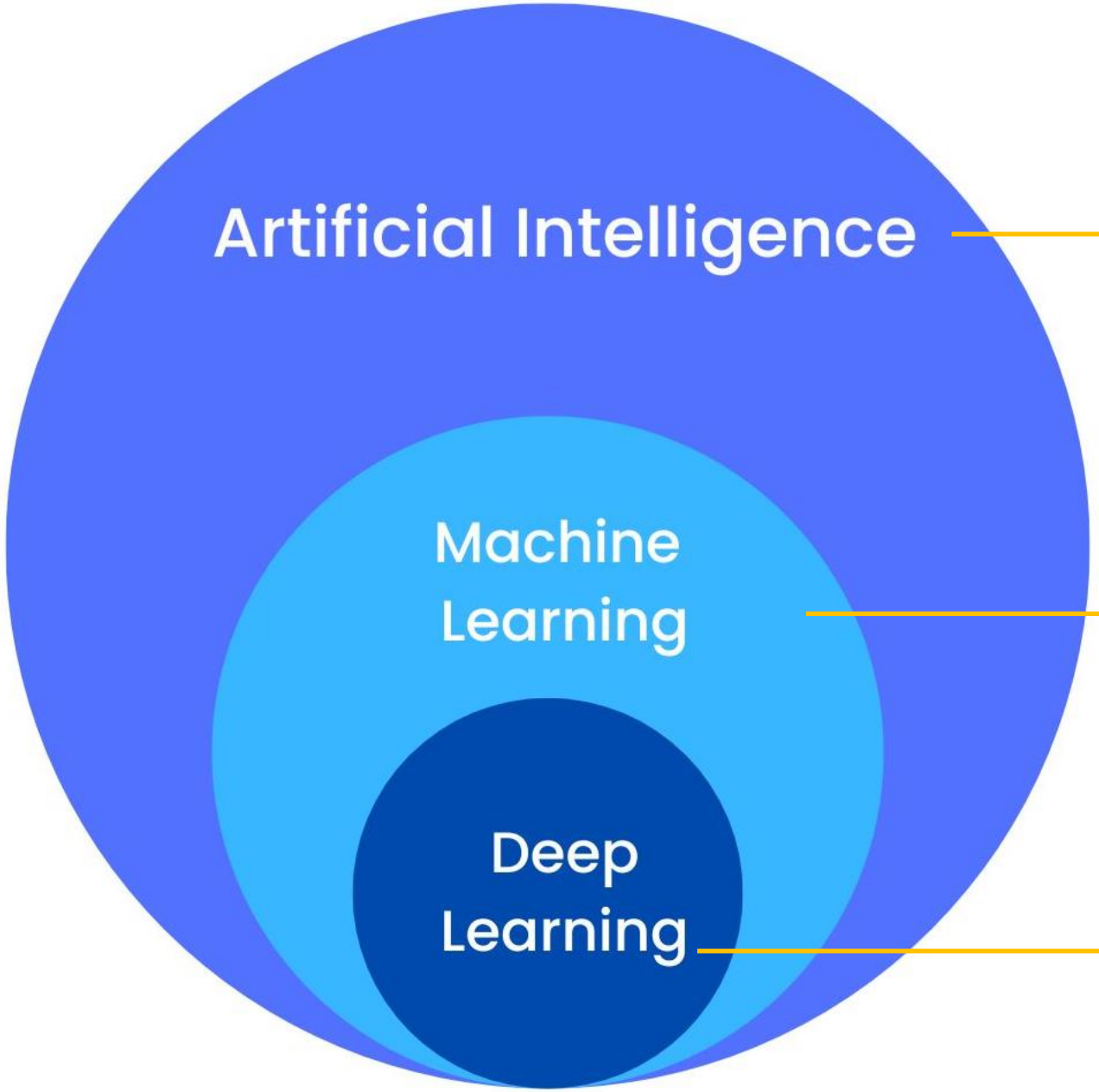
**Structured data**

| ID | Name | Age | Degree |
|---|---|---|---|
| 1 | John | 18 | B.Sc. |
| 2 | David | 31 | Ph.D. |
| 3 | Robert | 51 | Ph.D. |
| 4 | Rick | 26 | M.Sc. |
| 5 | Michael | 19 | B.Sc. |

# Types of data(cont.)

- **Structured Data** – Organized in rows and columns with a fixed schema. Examples: SQL databases, Excel spreadsheets, student marks, sales records, sensor readings.
- **Semi-structured Data** – Has some organization but no strict schema. Examples: JSON files, XML documents, emails, HTML pages, log files, NoSQL databases like MongoDB.
- **Unstructured Data** – No predefined format, hardest for AI to process directly. Examples: text documents, images, audio files, videos, PDFs, social media posts, MRI scans.

# AI vs Machine Learning vs Deep Learning

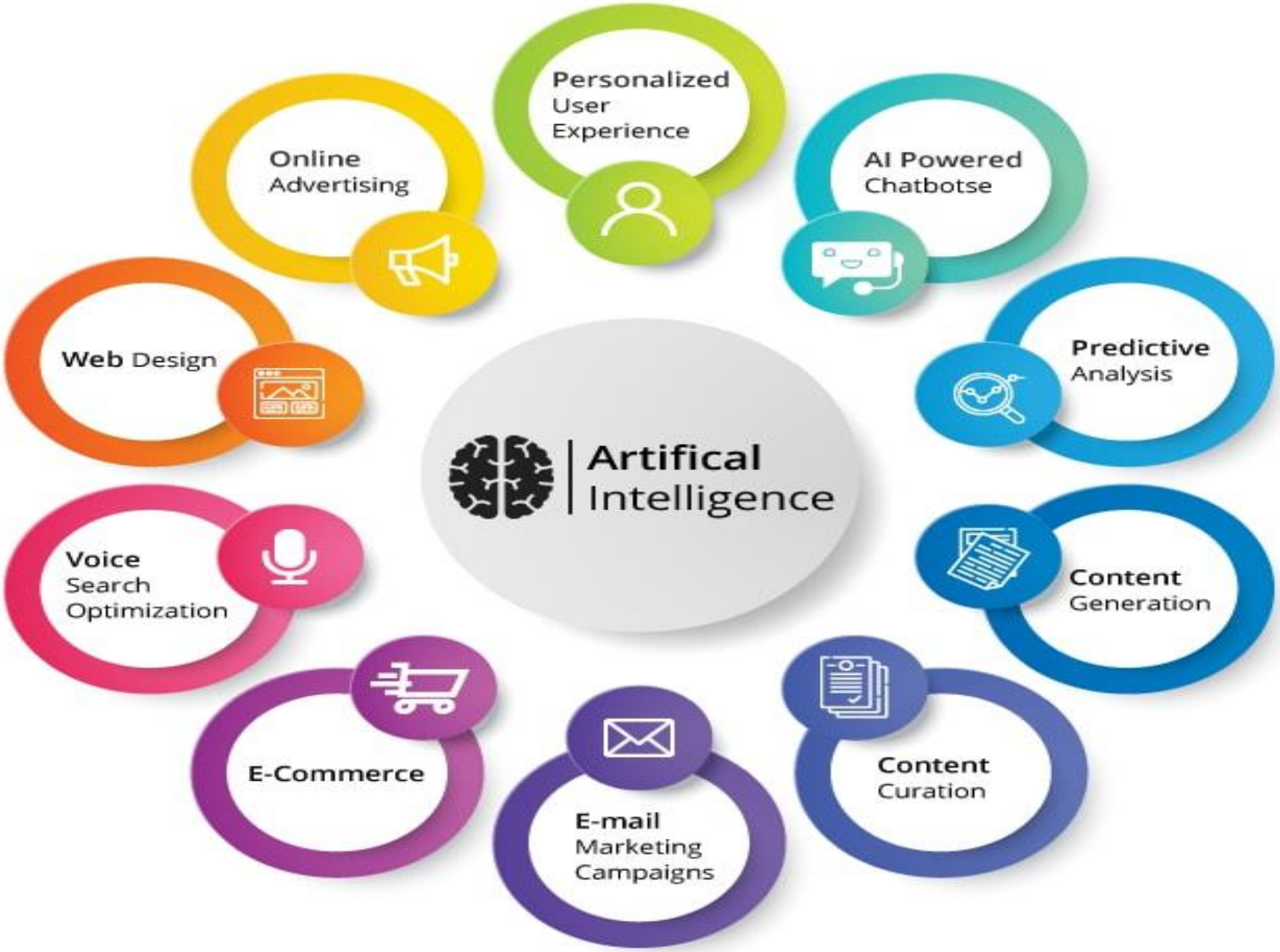



The broad science of making machines simulate human intelligence and decision-making.

A subset of AI where machines learn from data without being explicitly programmed.

Deep Learning is a subset of Machine Learning that mimics the human brain through layered neural networks.

# Applications of Artificial Intelligence


Artifical Intelligence
Personalized User Experience
AI Powered Chatbotse
Predictive Analysis
Content Generation
Content Curation
E-mail Marketing Campaigns
E-Commerce
Voice Search Optimization
Web Design
Online Advertising

# Advantages and Challenges of AI

# Advantages and challenges of AI

**Advantages:**

- Automates repetitive tasks → improves efficiency
- Fast data processing & decision-making
- High accuracy, reduces human errors
- 24/7 availability (no fatigue)
- Enables innovation (automation, smart systems)

**Challenges:**

- High development & maintenance cost
- Job displacement due to automation
- Data privacy & security concerns
- Bias and ethical issues in AI models
- Lack of human creativity & emotions

# Future of Artificial Intelligence

# Key trends shaping the future

- Smaller, more efficient models (on-device AI, edge computing)
- AI regulation and governance frameworks globally
- Explainable and trustworthy AI
- Human-AI collaboration over AI replacement
- Energy-efficient and sustainable AI infrastructure

# Identify AI vs non-AI applications

# Application list

- Swiggy predicting your delivery time based on traffic and weather
- A microwave set to heat food for 2 minutes
- Google Photos automatically grouping pictures by face
- A simple alarm clock that rings at 6:00 AM every day
- ChatGPT writing a paragraph based on your prompt
- An elevator moving to the floor when a button is pressed
- Grammarly suggesting better sentence structure in real time
- A light bulb that turns on when switched manually
- Spotify creating a personalized playlist every Monday

# Application list

- A vending machine dispensing a snack after coin insertion
- Tesla Autopilot detecting and avoiding obstacles on highway
- A barcode scanner reading product codes at billing counter
- Google Translate converting Hindi text to English instantly
- A petrol pump meter calculating fuel cost per litre
- Instagram reels showing content based on your scroll behaviour

## AI

- Swiggy predicting your delivery time based on traffic and weather
- Google Photos automatically grouping pictures by face
- ChatGPT writing a paragraph based on your prompt
- Grammarly suggesting better sentence structure in real time
- Spotify creating a personalized playlist every Monday
- Tesla Autopilot detecting and avoiding obstacles on highway
- Google Translate converting Hindi text to English instantly
- Instagram reels showing content based on your scroll behaviour

## Non-AI

- A microwave set to heat food for 2 minutes
- A simple alarm clock that rings at 6:00 AM every day
- An elevator moving to the floor when a button is pressed
- A light bulb that turns on when switched manually
- A vending machine dispensing a snack after coin insertion
- A barcode scanner reading product codes at billing counter
- A petrol pump meter calculating fuel cost per liter

रा.इ.सू.प्रौ.सं
NIELIT
BHUBANESWAR

# Thank you

Bijaylaxmi Behera

Faculty, NIELIT Bhubaneswar